# DEPARTMENT OF TECHNOLOGY

## Machine Learning

# Feature Scaling

**Learning Objective**

After this lecture, students will be able to:

- Understand the need for feature scaling.
- Identify when features have very different numerical ranges.
- Explain the basic idea of scaling features.
- Understand Standardization.
- Understand Min-Max Normalization.
- Distinguish between Standardization and Min-Max Normalization.
- Relate feature scaling to Gradient Descent.

## 1 Connection with Previous Topics

So far, we have studied:

- Simple Linear Regression
- Multiple Linear Regression
- Features and target variable
- Weights and bias
- Loss and cost function
- Gradient Descent

For Multiple Linear Regression, we have:

$$\boxed{\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_p x_p} \tag{1}$$

Gradient Descent updates the model parameters iteratively:

$$\boxed{w_j \leftarrow w_j - \alpha \frac{\partial J}{\partial w_j}} \tag{2}$$

where:

- $w_j$ = model parameter
- $\alpha$ = learning rate

- $J$ = cost function

Now consider the following question:

**Important**

**What happens if the input features have very different numerical ranges?**

## 2 Motivating Example

Consider a simple house price dataset.

| House | Area (sq.ft) | Bedrooms | Price (Lakhs) |
|---|---|---|---|
| H1 | 500 | 1 | 30 |
| H2 | 1000 | 2 | 50 |
| H3 | 1500 | 3 | 70 |
| H4 | 2000 | 4 | 90 |
| H5 | 2500 | 5 | 110 |

Here, the input features are:

$$x_1 = \text{Area}$$

and:

$$x_2 = \text{Bedrooms}$$

Observe the ranges:

$$500 \leq \text{Area} \leq 2500$$

while:

$$1 \leq \text{Bedrooms} \leq 5$$

Therefore, the two features have very different numerical magnitudes.

## 3 The Problem of Different Feature Scales

Consider the Multiple Linear Regression model:

$$\hat{y} = w_0 + w_1(\text{Area}) + w_2(\text{Bedrooms}) \tag{3}$$

The values of Area are in hundreds or thousands, whereas the values of Bedrooms are small. For example:

$$\text{Area} = 2000$$

but:

$$\text{Bedrooms} = 4$$

Both are valid features, but their numerical scales are very different.

This difference in scale can affect the optimization process when Gradient Descent is used. Recall:

$$w_j \leftarrow w_j - \alpha \frac{\partial J}{\partial w_j} \tag{4}$$

The magnitude of the feature values can influence the magnitude of the corresponding gradient.

Therefore, very different feature scales can make the optimization process less efficient.

$$\boxed{\text{Different Feature Scales} \rightarrow \text{Unequal Gradient Magnitudes} \rightarrow \text{Less Efficient Optimization}} \tag{5}$$

# 4 What is Feature Scaling?

**Feature Scaling** is the process of transforming numerical features so that they have comparable numerical scales.

For example, the original features may be:

$$\text{Area} = 500 \text{ to } 2500$$

and:

$$\text{Bedrooms} = 1 \text{ to } 5$$

After scaling, the features are represented using smaller and more comparable numerical values.

$$\boxed{\text{Original Features} \rightarrow \text{Scaling Transformation} \rightarrow \text{Scaled Features}} \tag{6}$$

# 5 What Does Scaling Actually Change?

Feature scaling changes the **numerical representation** of a feature.

It does not change the actual information represented by that feature.

For example, if one house has a larger area than another house, scaling does not change this relationship.

Before scaling:

$$500 < 1000 < 1500 < 2000 < 2500$$

After scaling, the ordering remains the same.

**Important**

Feature scaling changes the **scale of the values**, not the underlying information contained in the feature.

# 6 Common Feature Scaling Techniques

Two commonly used techniques are:

1. Standardization
2. Min-Max Normalization

# 7 Standardization

Standardization transforms a feature using its mean and standard deviation.

The formula is:

$$\boxed{z = \frac{x - \mu}{\sigma}} \tag{7}$$

where:

- $x$ = original feature value
- $\mu$ = mean of the feature
- $\sigma$ = standard deviation of the feature
- $z$ = standardized value

After standardization, the feature is centered around zero and has a standard deviation of approximately one.

$$\boxed{\text{Mean} \approx 0} \tag{8}$$

$$\boxed{\text{Standard Deviation} \approx 1} \tag{9}$$

# 8 Classroom Activity: Standardization

Consider the Area feature from the dataset:

$$500,\ 1000,\ 1500,\ 2000,\ 2500$$

Use the formula:

$$\boxed{z = \frac{x - \mu}{\sigma}}$$

Calculate:

1. The mean $\mu$ of the Area feature.
2. The standard deviation $\sigma$.
3. The standardized value of each observation.

Complete the following table:

| House | Area | Standardized Area |
|---|---|---|
| H1 | 500 | |
| H2 | 1000 | |
| H3 | 1500 | |
| H4 | 2000 | |
| H5 | 2500 | |

# 9 Understanding Standardized Values

After standardization:

- Values below the mean generally become negative.
- A value close to the mean becomes close to zero.
- Values above the mean generally become positive.

Therefore, standardized values indicate the position of an observation relative to the mean. For example:

$$z < 0$$

generally indicates a value below the mean, while:

$$z > 0$$

generally indicates a value above the mean.

# 10 Min-Max Normalization

Min-Max Normalization transforms a feature into a specified range.

The most common range is:

$$[0, 1]$$

The formula is:

$$\boxed{x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}} \tag{10}$$

where:

- $x$ = original value
- $x_{\min}$ = minimum value
- $x_{\max}$ = maximum value
- $x'$ = normalized value

# 11 Classroom Activity: Min-Max Normalization

Again consider the Area feature:

$$500,\ 1000,\ 1500,\ 2000,\ 2500$$

Identify:

$$x_{\min}$$

and:

$$x_{\max}$$

Then use:

$$\boxed{x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}}$$

Calculate the normalized value of each observation.
Complete the following table:

| House | Area | Normalized Area |
|---|---|---|
| H1 | 500 | |
| H2 | 1000 | |
| H3 | 1500 | |
| H4 | 2000 | |
| H5 | 2500 | |

# 12 Standardization vs Min-Max Normalization

| Property | Standardization | Min-Max Normalization |
|---|---|---|
| Main idea | Center and scale the feature | Map values to a fixed range |
| Formula | $z = \frac{x - \mu}{\sigma}$ | $x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$ |
| Typical result | Mean around 0 | Values between 0 and 1 |
| Uses | Commonly used when features need to be centered and scaled | Useful when a fixed range is desired |

# 13 Which Technique Should We Use?

There is no single scaling technique that is always best for every dataset.
The choice depends on:

- The nature of the data
- The machine learning algorithm
- The numerical range of the features

For now, remember the two basic approaches:

| Standardization and Min-Max Normalization |
|---|

# 14 Feature Scaling and Gradient Descent

The main connection with our previous topic is Gradient Descent.

We know:

$$w_j \leftarrow w_j - \alpha \frac{\partial J}{\partial w_j} \tag{11}$$

When features have very different scales, the optimization process can become inefficient. Scaling the features to comparable ranges can make the optimization process more balanced. Therefore:

| Feature Scaling → Comparable Feature Magnitudes → Better Optimization Behaviour |
|---|

(12)

This is particularly important when a model is trained using Gradient Descent.

# 15 Important Points to Remember

**Important**

1. Feature scaling changes the numerical scale of features.
2. It does not change the underlying information in the dataset.
3. Features with very different numerical ranges can affect Gradient Descent.
4. Standardization uses the mean and standard deviation.
5. Min-Max Normalization uses the minimum and maximum values.
6. Standardization generally produces values centered around zero.
7. Min-Max Normalization commonly produces values between 0 and 1.

# 16 Think and Discuss

1. Why do Area and Bedrooms have different feature scales?
2. Why can different feature scales affect Gradient Descent?
3. What is the purpose of feature scaling?
4. What is the difference between Standardization and Min-Max Normalization?
5. Does Standardization restrict values between 0 and 1?
6. What information does a standardized value represent?
7. Does feature scaling change the actual meaning of a feature?

# 17 Connection to the Next Topic

We have now learned how to prepare numerical features so that their scales are comparable.
But another question arises:

**Important**

**What if the relationship between an input feature and the target variable is not a straight-line relationship?**

For example, instead of:

$$\hat{y} = w_0 + w_1 x$$

can we allow the model to learn a curved relationship?
This leads to:

Polynomial Regression

Polynomial Regression will then help us understand:

Model Complexity → Underfitting → Overfitting → Regularization